

Touch Alarm

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This is a new type of touch alarm that uses an RF oscillator at its input. One special feature of this touch alarm is that it can use a big-size touch plate. Also, no shielded wire is required between the touch plate and the circuit.

Circuit and working

Fig. 1 shows the circuit of the touch alarm, which is built around CD4060 (IC1), timer 555 (IC2) and a few dis-

crete components. IC1 contains an easy-to-use astable or oscillator stage, followed by a 14-stage binary counter/divider.

Pulses from the oscillator stage are fed to the 14-bit binary counter stage. Not all the outputs of the binary counter are available externally. The frequency at pin 7 (Q3) of IC1 is actually divided by $2^4=16$ compared to the initial astable frequency. Similarly, Q4 will have half the frequency of Q3.

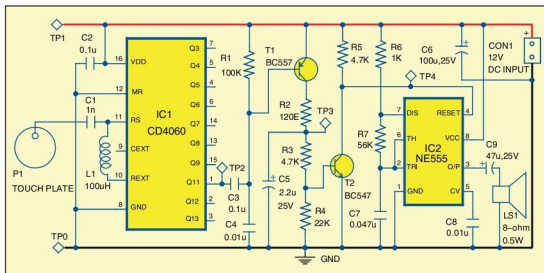


Fig. 1: Circuit of the touch alarm

Test Points

Test point	Details
TP0	0V, GND
TP1	+12V
TP2	1 kHz when touch plate not touched
TP3	Approx. 10V when touch plate not touched
TP4	0V when touch plate not touched

Reset pin (pin 12) is connected to ground to enable the pulses.

IC1 generates an astable frequency of around 4 MHz using low-cost inductor L1. The frequency available at pin 1 (Q11) of IC1 is around 1 kHz, which is

actually an astable frequency divided by 2^2 . This frequency keeps transistor T2 conducting. Pin 4 of IC2 is held low as long as transistor T2 conducts.

When you touch the touch plate, the oscillator stops working because of loading. No output signals are available at pin 1 of IC1, due to which transistor T2 stops conducting. When transistor T2

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